

Mining Various Kinds of Knowledge

- ☐ Multidimensional Data Summarization
- ☐ Mining Frequent Patterns, Associations, and Correlations
- ☐ Classification and Regression for Predictive Analysis
- ☐ Cluster Analysis
- ☐ Deep Learning
- ☐ Outlier Analysis
- ☐ Are All Mining Results Interesting?

Multidimensional Data Summarization

- ❑ Information integration and data warehouse construction
 - ❑ Data cleaning, transformation, integration, and multidimensional data model
- ❑ Data cube technology
 - ❑ Scalable methods for computing (i.e., materializing) multidimensional aggregates
 - ❑ OLAP (online analytical processing)
- ❑ Multidimensional concept description: Characterization and discrimination
 - ❑ Generalize, summarize, and contrast data characteristics, e.g., dry vs. wet region



Image Source:
Artificial Intelligence in Daily Life by Raymond S. T. Lee

Pattern Discovery: Mining Frequent Patterns, Associations, and Correlations

- Frequent patterns (or frequent itemsets)
 - What items are frequently purchased together in your Walmart?
- Association and Correlation Analysis

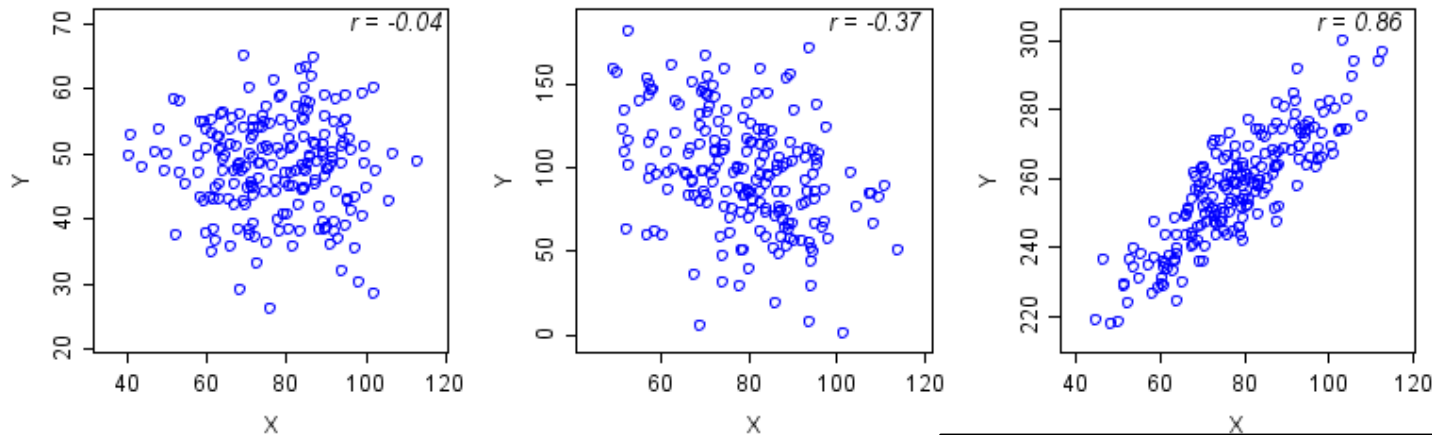


Image Source: [Data Analysis from Boston University](#)

- A typical association rule
 - Diaper $\rightarrow$ Beer [0.5%, 75%] (support, confidence)
 - Are strongly associated items also strongly correlated?
- How to mine such patterns and rules efficiently in large datasets?
- How to use such patterns for classification, clustering, and other applications?

Classification and Regression for Predictive Analysis

❑ Classification and label prediction

- ❑ Construct models (functions) based on some training examples
- ❑ Describe and distinguish classes or concepts for future prediction
 - ❑ Ex. 1. Classify countries based on (climate)
 - ❑ Ex. 2. Classify cars based on (gas mileage)
- ❑ Predict some unknown class labels

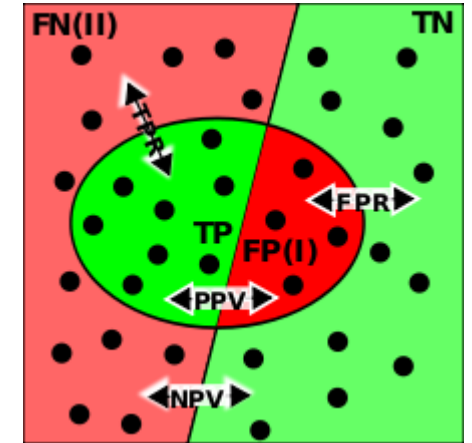


Image Source: [Binary Classification, Wikipedia](#)

❑ Typical methods

- ❑ Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, ...

❑ Typical applications:

- ❑ Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, ...